

and gynaecologic surgical procedures. The system was named after the eminent Italian polymath Leonardo da Vinci, courtesy his invention of a robot, the “mechanical knight”. It was a humanoid automaton, believed to have been designed and constructed by the artist himself in 1495. The teeny-weeny instruments in the aforementioned surgical system move like a human hand but with a wider range of motion. The minute size of the toolkit makes it simple for surgeons to smoothly operate through a single or a very few minor incisions.

India picture

Statistics prove that at present, India boasts of 66 robotic surgery centers with 71 robotic systems already installed to



implement the surgical procedures (confirmed by ClinicSpots holistic healthcare). Some of the top robotic surgery hospitals in India are Apollo Hospitals, Delhi, Hinduja Hospital, Mumbai, Global Hospitals Group India, Fortis Hospital, Delhi, Max Hospital, New Delhi, Medanta Hospital, Gurugram, ILS Hospitals, Kolkata.

It is true that India is steadily embracing AI-aided robotic surgeries and the number of robot-assisted surgery is certainly increasing within the country because of its elevated accuracy levels and highly rewarding outcomes.

In India, robot-assisted surgeries have been conducted for more than two decades now, primarily utilizing the Da Vinci system. More recently, machines by Medtronic and Stryker are also introduced.

Is robotic surgery risk-prone?

When enquired if these surgeries are susceptible to high risks and what happens if a robot suffers a technical



breakdown in the middle of a surgery, to that, Dr. Subhash Dhiware, consultant, orthopaedics department, KDAH, assures that “there is no extra risk involved in robot-backed surgeries. If the robot hits a snag, the surgeon present around can immediately take over and complete the remaining job in standard fashion. But chances of such emergencies and unpredictable circumstances are extremely low as every piece of equipment is tested before initiating a surgery”.

“All surgical procedures are risk-prone to some degree, including robotic surgeries”, Dr. Rayapudi accepts this fact. “However, in the event of any issue, say a technical failure in the middle of a surgery, a team of surgeons is present to take over from the robotic system and intervene if necessary,” he reminds to retain faith in the technological breakthrough in medical science that is under discussion.

Robotic revolution in medical science

Robot-assisted surgery has a proven track record for general surgery, gynaecology, urology, cardiac, H&N (head and neck), oncology, GI (Gastrointestinal), transplant, vascular and thoracic procedures, and so on. Apart from Da Vinci, other robots are unleashing disruptive technology even in cardiology, orthopaedic joint replacement and lung biopsy. Robotics with an impressive success ratio in surgical procedures could unlock its significant value and is



definitely here to stay with a promise of greater evolution in the coming days.

Can robots replace surgeons?

It is important to note that a highly dedicated team of remote surgeons and AI experts play a pivotal role in a robot-aided surgery. They coordinate to design and strategise the surgery using superior imaging techniques and computer simulations. They can use this material to create a detailed surgical plan that takes into account the patient’s unique anatomy and any potential complications. During the surgery, distant surgeons and AI exponents can provide real-time support to the on-site surgical team. They can use sensors to monitor the patient’s vital signs, provide guidance on the use of the robotic system and make adjustments to the surgical plan as and when required. After the surgery, this crucial data can be utilised to optimize the patient’s health and well-being and ensure to address any issue or complication instantly.



Robotic surgery as of now is not autonomous or fully automated. The robot does not perform the surgery on its own but serves as an extension of the surgeon’s hands, allowing more correct and controlled movements. The surgeon maneuvers the procedure and makes all necessary decisions during the surgery, leaving the robot to obey the surgeon’s commands.

In robot-assisted surgeries, one surgeon typically operates one robotic system throughout the duration of a single surgery, regardless of how long it takes. This allows the surgeon to concentrate solely on the procedure and utilize the robot’s extraordinary accuracy and meticulousness to perform the surgery. **d**